

Introduction to Machine Learning
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Lecturer: Zhihui Li, Xiaojun Chang
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Name: Jinghao Chen

Homework 3
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ID: PB23010359

Notice, to get the full credits, please present your solutions step by step.

Exercise 1: Affine Sets and Projections

- (1) Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and consider

$$U = \{x \in \mathbb{R}^n : Ax = b\}.$$

- (a) Show that U is an affine set. Identify its direction space in terms of A .
(b) Prove that for any $x_0 \in U$,

$$U = x_0 + \mathcal{N}(A),$$

where $\mathcal{N}(A)$ denotes the null space of A .

- (2) Suppose $A \in \mathbb{R}^{m \times n}$ has full row rank and $U = \{x : Ax = b\}$ is nonempty.

- (a) Consider the projection of a point $y \in \mathbb{R}^n$ onto U :

$$p = \underset{x \in U}{\mathbf{argmin}} \|x - y\|_2^2.$$

Use Lagrange multipliers to derive the formula

$$p = y - A^\top (AA^\top)^{-1} (Ay - b).$$

- (b) In $\mathbb{R}^3$, let

$$U = \{x \in \mathbb{R}^3 : x_1 + 2x_2 - x_3 = 3\}, \quad y = (1, 0, 0)^\top.$$

Compute explicitly the projection p of y onto U .

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Solution 1: Affine Sets and Projections

(1) (a) $\forall x_1, x_2 \in U, \forall \theta \in \mathbb{R},$

$$A(\theta x_1 + (1-\theta)x_2) = \theta Ax_1 + (1-\theta)Ax_2 = \theta b + (1-\theta)b = b \implies \theta x_1 + (1-\theta)x_2 \in U.$$

namely U is an affine set.

$$\forall x \in U, \exists y \in \mathbb{R}^n, \text{ s.t.}$$

$$x + y \in U \iff A(x + y) = Ax + Ay = b + Ay = b \iff Ay = 0 \iff y \in \mathcal{N}(A)$$

namely the direction space of U is $\mathcal{N}(A)$.

(b) $\forall x_0 \in U,$

$$x \in U \iff Ax = b \iff A(x - x_0) = 0 \iff x - x_0 \in \mathcal{N}(A) \iff x \in x_0 + \mathcal{N}(A)$$

Thus, $U = x_0 + \mathcal{N}(A)$.

(2) (a)

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \|x - y\|_2^2 \\ \text{s.t.} \quad & Ax = b \end{aligned}$$

$$\xrightarrow{\text{def}} L(x, \lambda) = \|x - y\|_2^2 + \lambda^\top (Ax - b)$$

$$\nabla_x L = 2(x - y) + A^\top \lambda \xrightarrow{\nabla_x L = 0} p = x^* = y - \frac{1}{2} A^\top \lambda$$

$$Ax = b \implies A(y - \frac{1}{2} A^\top \lambda) = b \implies AA^\top \lambda = 2(Ay - b)$$

A has full row rank $\implies AA^\top \in \mathbb{R}^{m \times m}$ is symmetric positive definite. $\implies AA^\top$ is invertible.

$$\implies \lambda = 2(AA^\top)^{-1}(Ay - b) \implies p = y - A^\top (AA^\top)^{-1}(Ay - b)$$

(b)

$$A = (1, 2, -1), \quad b = 3, \quad y = (1, 0, 0)^\top.$$

$$AA^\top = 1^2 + 2^2 + (-1)^2 = 6, \quad Ay - b = 1 * 1 + 2 * 0 + (-1) * 0 - 3 = -2.$$

$$p = y - A^\top (AA^\top)^{-1}(Ay - b) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \cdot \frac{1}{6} \cdot (-2) = \begin{pmatrix} \frac{4}{3} \\ \frac{2}{3} \\ -\frac{1}{3} \end{pmatrix}.$$

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Exercise 2: Convex Sets, Images and Preimages

Let $C \subseteq \mathbb{R}^n$ be a nonempty convex set and let $A \in \mathbb{R}^{m \times n}$, $a \in \mathbb{R}^m$.

- (1) Show that the affine image of C ,

$$D = \{y \in \mathbb{R}^m : y = Ax + a, x \in C\},$$

is convex.

- (2) Define the affine preimage of C by

$$E = \{y \in \mathbb{R}^m : Ay + a \in C\}.$$

Show that E is convex.

- (3) (Epigraph and hypograph)

- (a) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. The epigraph of f is

$$\mathbf{epi} f = \{(x, t) \in \mathbb{R}^{n+1} : t \geq f(x)\}.$$

Show that $\mathbf{epi} f$ is a convex set.

- (b) Let $g : \mathbb{R}^n \rightarrow \mathbb{R}$ be concave. Its hypograph is

$$\mathbf{hypo} g = \{(x, t) \in \mathbb{R}^{n+1} : t \leq g(x)\}.$$

Show that $\mathbf{hypo} g$ is convex.

- (4) (Interior and relative interior – computation) You do not need to prove your answers, but you should explain them briefly.

- (a) Find $\mathbf{int} C$ and $\mathbf{relint} C$ for

$$C = \{x \in \mathbb{R}^3 : x_1^2 + x_2^2 < 4, x_3 \geq 0\}.$$

- (b) Let

$$K = \{x \in \mathbb{R}^n : x_i \geq 0, \sum_{i=1}^n x_i = 1\}$$

be the standard probability simplex. Determine $\mathbf{int} K$ (as a subset of $\mathbb{R}^n$) and $\mathbf{relint} K$.

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Solution 2: Convex Sets, Images and Preimages

$$(1) \quad \forall y_1, y_2 \in D, \exists x_1, x_2 \in C, \text{ s.t. } y_1 = Ax_1 + a, y_2 = Ax_2 + a.$$

$$C \text{ is convex} \implies \forall \theta \in [0, 1], \theta x_1 + (1 - \theta)x_2 \in C,$$

$$\implies \theta y_1 + (1 - \theta)y_2 = \theta(Ax_1 + a) + (1 - \theta)(Ax_2 + a) = A(\theta x_1 + (1 - \theta)x_2) + a \in D$$

namely D is convex.

$$(2) \quad \forall y_1, y_2 \in E, Ay_1 + a \in C, Ay_2 + a \in C,$$

$$C \text{ is convex.} \implies \forall \theta \in [0, 1], \theta(Ay_1 + a) + (1 - \theta)(Ay_2 + a) = A(\theta y_1 + (1 - \theta)y_2) + a \in C$$

$$\implies \theta y_1 + (1 - \theta)y_2 \in E, \text{ namely } E \text{ is convex.}$$

$$(3) \quad (a) \quad \forall (x_1, t_1), (x_2, t_2) \in \mathbf{epi} f, t_1 \geq f(x_1), t_2 \geq f(x_2),$$

$$\begin{aligned} f \text{ is convex} \implies \forall \theta \in [0, 1], \theta t_1 + (1 - \theta)t_2 &\geq \theta f(x_1) + (1 - \theta)f(x_2) \\ &\geq f(\theta x_1 + (1 - \theta)x_2) \end{aligned}$$

$$\implies (\theta x_1 + (1 - \theta)x_2, \theta t_1 + (1 - \theta)t_2) \in \mathbf{epi} f, \text{ namely } \mathbf{epi} f \text{ is convex.}$$

$$(b) \quad \textcircled{1} \quad \forall (x_1, t_1), (x_2, t_2) \in \text{hypo } g, t_1 \leq g(x_1), t_2 \leq g(x_2),$$

$$\begin{aligned} g \text{ is concave} \implies \forall \theta \in [0, 1], \theta t_1 + (1 - \theta)t_2 &\leq \theta g(x_1) + (1 - \theta)g(x_2) \\ &\leq g(\theta x_1 + (1 - \theta)x_2) \end{aligned}$$

$$\implies (\theta x_1 + (1 - \theta)x_2, \theta t_1 + (1 - \theta)t_2) \in \text{hypo } g, \text{ namely } \text{hypo } g \text{ is convex.}$$

$\textcircled{2}$ Let $h = -g$, then h is convex.

$$t \leq g(x) \iff -t \geq -g(x) = h(x)$$

$$(x, t) \in \text{hypo } g \iff (x, -t) \in \text{epi } h$$

Define the linear mapping:

$$T : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{n+1}, \quad T \begin{pmatrix} x \\ t \end{pmatrix} = \begin{pmatrix} I_n & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix} = \begin{pmatrix} x \\ -t \end{pmatrix}$$

According to (1) and (3)(a), $\mathbf{epi} h$ is convex $\implies T(\text{hypo } g) = \mathbf{epi} h$ is convex $\implies \text{hypo } g$ is convex.

$$(4) \quad (a)$$

$$\mathbf{int} C = \mathbf{relint} C = \{x \in \mathbb{R}^3 : x_1^2 + x_2^2 < 4, x_3 > 0\},$$

Proof:

$$B((0, 0, 1), \frac{1}{2}) \subset C \implies 3 \geq \dim C \geq \dim B((0, 0, 1), \frac{1}{2}) = 3$$

$$\implies \dim C = \dim(\mathbf{aff} C) = 3 \implies \mathbf{int} C = \mathbf{relint} C$$

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(b)

$$\mathbf{int} K = \emptyset, \quad \mathbf{relint} K = \{x \in \mathbb{R}^n : x_i > 0, \sum_{i=1}^n x_i = 1\}.$$

Proof:

$$\dim K = n - 1 < n \implies \mathbf{int} K = \emptyset$$

$$\mathbf{aff} K = \{x \in \mathbb{R}^n : \sum_{i=1}^n x_i = 1\} \implies \mathbf{relint} K = \{x \in \mathbb{R}^n : x_i > 0, \sum_{i=1}^n x_i = 1\}.$$

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Exercise 3: Cones and Farkas-type Results

Let $A = (a_1, \dots, a_n) \in \mathbb{R}^{m \times n}$ and consider the conic hull

$$\text{cone}(A) = \left\{ \sum_{i=1}^n \alpha_i a_i : \alpha_i \geq 0 \right\} \subseteq \mathbb{R}^m.$$

- (1) Show that $\text{cone}(A)$ is a convex cone, i.e.,

$$x, y \in \text{cone}(A), \lambda, \mu \geq 0 \Rightarrow \lambda x + \mu y \in \text{cone}(A).$$

- (2) Show that $\text{cone}(A)$ is closed.

- (3) (Gordan's theorem) Let $A \in \mathbb{R}^{m \times n}$. Prove that exactly one of the following two systems has a solution:

- There exists $x \in \mathbb{R}^n$ such that $Ax > 0$ (componentwise).
- There exists $y \in \mathbb{R}^m$, $y \geq 0$, $y \neq 0$, such that $A^\top y = 0$.

Hint: Consider appropriate cones and use separation theorems (you may assume standard separation results for closed convex cones).

- (4) Show how Farkas' Lemma can be deduced from Gordan's theorem in part (3).

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Solution 3: Cones and Farkas-type Results

- (1) $\forall x, y \in \text{cone}(A), \exists \alpha_i, \beta_i \geq 0, \text{ s.t. } x = \sum_{i=1}^n \alpha_i a_i, \quad y = \sum_{i=1}^n \beta_i a_i.$

$$\begin{aligned} \forall \lambda, \mu \geq 0, \lambda \alpha_i + \mu \beta_i \geq 0, &\implies \lambda x + \mu y = \lambda \sum_{i=1}^n \alpha_i a_i + \mu \sum_{i=1}^n \beta_i a_i \\ &= \sum_{i=1}^n (\lambda \alpha_i + \mu \beta_i) a_i \in \text{cone}(A) \end{aligned}$$

- (2) First, prove that for each linear independent index set $I \subseteq \{1, 2, \dots, n\}$, the conic hull $\text{cone}(A_I)$ is closed, where $A_I = (a_i)_{i \in I}$.

$\forall \{x_k\} \subset \text{cone}(A_I)$ with $x_k \rightarrow x \in \mathbb{R}^m$.

$$x_k = \sum_{i \in I} \alpha_i^{(k)} a_i, \quad \alpha_i^{(k)} \geq 0.$$

If $\{\alpha_i^{(k)}\}$ is unbounded, then $\exists$ subsequence $\{\alpha_i^{(k_j)}\}$, s.t. $\|\alpha^{(k_j)}\| \rightarrow \infty$. Define the normalized sequence: $\beta^{(k_j)} = \frac{\alpha^{(k_j)}}{\|\alpha^{(k_j)}\|}$. Since $\beta^{(k_j)}$ is bounded, $\exists$ convergent subsequence $\beta^{(k_{j_l})} \rightarrow \beta \geq 0, \|\beta\| = 1$. According to the continuity of linear mapping, $A_I \beta^{(k_{j_l})} \rightarrow A_I \beta$. On the other hand, $A_I \beta^{(k_{j_l})} = \frac{x_{k_{j_l}}}{\|\alpha^{(k_{j_l})}\|} \rightarrow 0$. Thus, $A_I \beta = 0$, i.e.,

$\sum_{i \in I} \beta_i a_i = 0$, which contradicts the linear independence of $\{a_i\}_{i \in I}$.

Therefore, $\{\alpha_i^{(k)}\}$ is bounded $\implies \exists$ convergent subsequence $\alpha^{(k_j)} \rightarrow \alpha \geq 0$. According to the continuity of linear mapping, $\lim_{j \rightarrow \infty} A_I \alpha^{(k_j)} = A_I \alpha \in \text{cone}(A_I) \implies x \in \text{cone}(A_I) \implies \text{cone}(A_I)$ is closed.

Second, every point in C can be expressed as a nonnegative combination of at most m linearly independent columns of A . Therefore, there are only finite linear independent index sets. For each linear independent index set I , $\text{cone}(A_I)$ is closed $\implies \text{cone}(A) = \bigcup_I \text{cone}(A_I)$ is closed.

- (3) On the one hand, assume the first one has no solution. Then the sets

$$U = \{Ax : x \in \mathbb{R}^n\}, \quad V = \{y \in \mathbb{R}^m : y > 0\}$$

are disjoint convex sets. By the separation theorem for convex sets, $\exists a \neq 0 \in \mathbb{R}^m$ and $\alpha \in \mathbb{R}$, s.t.

$$a^\top u \leq \alpha, \forall u \in U, \quad a^\top v \geq \alpha, \forall v \in V$$

$$0 \in U, \quad a^\top 0 = 0 \leq \alpha \implies \alpha \geq 0. \quad \forall v \in V, \text{ let } v \rightarrow 0, \text{ then } \alpha \leq a^\top v \rightarrow 0 \implies \alpha = 0.$$

Besides,

$$“u \in U \iff -u \in U” \implies a^\top u \leq 0, a^\top (-u) \leq 0 \implies a^\top u = 0, \forall u \in U$$

$$\left. \begin{aligned} a^\top u &= a^\top Ax = 0, \forall x \in \mathbb{R}^n \implies A^\top a = 0 \\ a^\top v &\geq 0, \forall v \in V \implies a \geq 0 \end{aligned} \right\} \implies \exists y = a \geq 0, y \neq 0, \text{ s.t. } A^\top y = 0.$$

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On the other hand, assume the second one holds. Then

$$\exists y \in \mathbb{R}^m, y \neq 0, y \geq 0, \text{ s.t. } A^\top y = 0. \implies \forall x \in \mathbb{R}^n, \quad y^\top Ax = x^\top A^\top y = 0$$

Because $y \neq 0$ and $y \geq 0$, $\exists i$, s.t. $y_i > 0 \implies Ax > 0$ is impossible. Thus, the first one has no solution.

In conclusion, the first one and the second one are mutually exclusive and collectively exhaustive, i.e., exactly one of the two systems has a solution.

- (4) Farkas' Lemma: Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Then exactly one of the following two systems has a solution:

- There exists $x \in \mathbb{R}^n$ such that $Ax = b, x \geq 0$ (componentwise).
- There exists $y \in \mathbb{R}^m$ such that $A^\top y \geq 0, b^\top y < 0$.

Deduce Farkas' Lemma from Gordan's theorem:

Construct the augmented matrix $\tilde{A} = [A, -b]^\top \in \mathbb{R}^{(n+1) \times m}$.

Case 1: $\exists y \in \mathbb{R}^m$, s.t. $\tilde{A}y > 0 \iff A^\top y > 0, -b^\top y > 0 \iff A^\top y > 0, b^\top y < 0 \implies$ the second one in Farkas' Lemma has a solution.

Case 2: $\exists v \in \mathbb{R}^{n+1}, v \geq 0, v \neq 0$, s.t. $\tilde{A}^\top v = 0$. Define $v = (x, t)$, $x \in \mathbb{R}^n, t \in \mathbb{R}$. Then

$$\tilde{A}v = Ax - bt = 0, (x, t) \geq 0, (x, t) \neq 0$$

If $t > 0$, let $x' = \frac{x}{t}$, then

$$Ax' = b, \quad x' \geq 0,$$

namely the first one in Farkas' Lemma has a solution.

If $t = 0$, then $Ax = 0, x \geq 0, x \neq 0$. If the first one has solution, it does not matter. If the first one has no solution, it means $b \notin \text{cone}(A)$. According to (1) and (2), $\text{cone}(A)$ is convex and closed. By the separation theorem for closed convex sets, $\exists y \neq 0 \in \mathbb{R}^m$ and $\alpha \in \mathbb{R}$, s.t.

$$y^\top u \geq \alpha, \forall u \in \text{cone}(A), \quad y^\top b < \alpha$$

$0 \in \text{cone}(A) \implies \alpha \leq 0$. If $\exists u_0 \in \text{cone}(A)$, s.t. $y^\top u_0 < 0$, then for $\forall \lambda > 0$, $\lambda u_0 \in \text{cone}(A)$, $y^\top(\lambda u_0) = \lambda(y^\top u_0) \rightarrow -\infty$, which contradicts $y^\top u \geq \alpha$. Thus,

$$y^\top u \geq 0, \forall u \in \text{cone}(A), \quad y^\top b < \alpha \leq 0$$

$$a_i = Ae_i \in \text{cone}(A) \implies y^\top a_i \geq 0, \forall i \implies A^\top y \geq 0, \quad b^\top y = y^\top b < 0$$

namely the second one in Farkas' Lemma has a solution. ■